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RESEARCH

Spontaneous Nystagmus Violating the Alexander's Law: Neural Substrates and Mechanisms

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Abstract

Alexander's law states that spontaneous nystagmus increases when looking in the direction of fast-phase and decreases during gaze in slow-phase direction. Disobedience to Alexander's law is occasionally observed in central nystagmus, but the underlying neural circuit mechanisms are poorly understood. In a retrospective analysis of 2,652 patients with posterior circulations stroke, we found a violation of Alexander's law in one or both directions of lateral gaze in 17 patients with lesions of unilateral lateral medulla affecting the vestibular nucleus. Patients with vestibular neuritis served as a control. When Alexander's law is violated, the time constant (Tc) was larger than that in the controls (median [interquartile range, IQR]: 14.4s [6.4–38.9] vs 9.0s [IQR 5.5–12.6], $p=0.036$) while the Tc did not differ between the groups when Alexander' law is obeyed (9.6s [3.6–16.1] vs 9.0s [5.5–12.6], $p=0.924$). To test the study hypothesis that an unstable neural integrator may generate nystagmus violating Alexander's law, we utilized the gaze-holding neural integrator model incorporating brainstem leaky neural integrator and negative velocity feedback loop via the cerebellum. The lesion-induced changes included false rotational cue, primarily attributed to central vestibular imbalance, and unstable neural integrator, examined in two ways: hyperexcitable brainstem neural integrator and paradoxical excitatory effect of Purkinje cells. With normal integrator function, the false rotational cue generated nystagmus consistent with Alexander's law. However, both types of unstable neural integrators tested produced nystagmus that violated Alexander's law. We propose that when the neural integrator is unstable with lesions in the brainstem neural integrator itself or the neural synapse between Purkinje cells and the brainstem vestibular nucleus, nystagmus violates Alexander's law. The spontaneous nystagmus violating Alexander's law may be the useful clue for identifying central vestibular syndrome.

Keywords Nystagmus · Alexander's law · Neural integrator · Lateral medullary infarction · Cerebellum

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Introduction

Alexander's law refer to the phenomenon that spontaneous nystagmus increases during the gaze in the fast-phase direction and decreases during the gaze in the opposite slow-phase direction [1]. This law applies to most nystagmus from either peripheral or central lesions [2, 3]. A few hypotheses have been proposed to explain this behavior, which include adaptive changes in the function of velocity-to-position neural integrator [4, 5] and inherent physiological properties of the brainstem nuclei that process information both for the vestibulo-ocular reflex (VOR) and normal gaze holding [6, 7].

Previously, spontaneous nystagmus that contravenes the Alexander's law has been described in a patient with upbeat nystagmus associated with Wernicke's encephalopathy [8]. Although the exact mechanism behind this phenomenon remains uncertain, one plausible explanation is an unstable neural integrator, as suggested by the slow phases of nystagmus with an exponentially increasing velocity [8]. In this context, we hypothesized that coalescence of unstable neural integrator with direction-fixed spontaneous nystagmus may give rise to violation of the Alexander's law.

This study aimed to define the characteristics and anatomical substrates of spontaneous nystagmus that violates the Alexander's law, and elucidate the mechanism using quantitative oculographic analyses and a mathematical model of an unstable neural integrator.

Table 1 Demographic and clinical characteristics of the patients

	Patients with nystagmus violating Alexander's law ($n=17$)	Patients with vestibular neuritis ($n=20$)	*p*
Male sex, n (%)	9 (53)	13 (65)	0.272
Age, y, mean (SD)	62 (13)	56 (10)	0.149
Lesion side, right, n (%)	9 (53)	14 (70)	0.286
Onset to evaluation, day, median (IQR)	1 (0.5–2.5)	1 (1–1)	0.478
SN, °/s, median (IQR)	3.3 (2.0-7.8)	4.8 (2.5–6.5)	0.557
Abnormal vHIT gain for HC, n (%)	6 (43)	20 (100)	<0.001
Ipsilesionally decreased	2	20	
Bilaterally decreased	1	0	
Contralesionally increased	2	0	
Bilaterally increased	1	0	

HC horizontal canal, *vHITs* video head impulse tests, *IQR* interquartile range, *SN* spontaneous nystagmus

Materials and Methods

Subjects

We retrospectively reviewed the medical records 2,652 patients with posterior circulations stroke (PCS) in the stroke registry at two University Hospitals in South Korea, Pusan National University and Pusan National University Yangsan Hospitals during a 13-year period. Among 239 patients with spontaneous horizontal or mixed horizontal and torsional nystagmus while fixating on a target straight ahead, we found 17 patients (7%) having the horizontal nystagmus that disobeyed the Alexander's law, i.e., decrease of nystagmus when looking in the direction of fast-phases and/or increase of nystagmus during the gaze in the direction of slow-phases (Video 1). The patients included nine men with the age ranging from 31 to 87 years (mean ± SD = 62 ± 13). For comparison, 20 patients with acute vestibular neuritis (VN, 13 men, 56 ± 10 years) served as a control. The demographic and clinical characteristics of the subjects are summarized in Table 1.

Clinical and Radiological Evaluation

All patients received structured neurological and neuro-otological examination by the authors (J-H.C., K.D.C). Bedside neuro-otological evaluation included 4-item HINTS plus (Head Impulse, Nystagmus, Test of Skew, and acute hearing loss detected by finger rubbing) [9]. All patients had evaluation within 7 days from symptom onset. Patients received stroke protocol MRIs including axial T2, fluid-attenuated inversion recovery image, DWI, and angiography.

Eye Movement Recording and Analyses

Eye movements were recorded using three-dimensional video-oculography (VOG, SLVNG, SLMED, Seoul, South Korea) with the patients seated upright. The VOG device was calibrated for horizontal and vertical eye positions using its calibration system with laser projections on the screen at a distance of 1.5 m. After calibration, we recorded nystagmus by instructing the patients to look at a target straight-ahead and displaced ± 20° horizontally with lights on. The target was present for 20 s at each position.

We analyzed the nystagmus at three gaze positions: the primary and each eccentric gaze. The horizontal eye position was coded positive when looking in the fast-phase direction of spontaneous nystagmus. Eye position signals were differentiated, and the slow-phase velocities (SPVs) of nystagmus were calculated after removing fast phases using a MATLAB script. An average value of all SPVs was measured at the primary and each eccentric gaze position. To determine an obedience of Alexander's law, we applied a linear regression through all SPV data points at the three gaze direction as previously described [7, 10]. Then, we measured the slope of average SPVs between the primary and eccentric gaze positions in the fast- and the slow-phase directions [3, 10, 11]. When the slopes of both fast and slow phase directions were positive, we defined that the nystagmus obeyed the Alexander's law. In contrast, we defined that the nystagmus violated the Alexander's law when at least one of the slopes was negative. We also calculated the time constant (Tc) of nystagmus at each eccentric gaze position as follows [2, 3, 10, 11].

$$Tc = \Delta E_{pos}/\Delta E_{vel}$$

ΔE_{pos}: difference of eye position between the two positions.

ΔE_{vel}: difference of slow-phase velocity between the two positions.

We also recorded spontaneous nystagmus without visual fixation, and positional nystagmus induced by various positional maneuvers including lying down from sitting, head turning to either side while supine, straight-head hanging, and Dix-Hallpike maneuver to each side.

Laboratory Evaluation

The VOR during head impulses was quantitatively assessed in 14 patients using a video-based equipment (SLMED, Seoul, Korea) as described previously [12]. The VOR gain was calculated as the ratio of the area under the entire eye-velocity relative to the area under the entire head-velocity. Reference data were obtained from 31 normal controls (normal gains for the horizontal canals = 0.85–1.02, for the anterior canals = 0.91–1.06, for the posterior canals = 0.88–1.05). The subjective visual vertical (SVV) tilt was measured in 13 patients by seating them upright in a dark room and asking to align a rod (80 cm long and 0.3 cm wide) vertically. The rod was presented randomly at various angles from the vertical at a distance of 130 cm from the patient's eyes. The SVV tilt was determined by calculating the average deviation from the earth vertical during five adjustments. The SVV tilt was considered abnormal when it exceeded the normal values obtained from healthy controls (-3.0° to 3.0° in both eyes; a negative value indicates a counterclockwise rotation) [12].

Statistical Analysis

All analyses were performed with SPSS (version 22.0, Chicago, IL, USA). Continuous variables were compared using the t test or Mann–Whitney U test, and nominal variables were compared with the $\chi 2$ or Fisher exact tests. The significance level was set at $p < 0.05$.

Results

Clinical and Radiological Characteristics

During visual fixation, patients had mixed horizontal-torsional-upbeat ($n = 11$), pure horizontal ($n = 4$), and mixed horizontal-torsional ($n = 2$) nystagmus (Table 2). The direction of horizontal nystagmus was contralesional in all. Without visual fixation, all patients showed mainly horizontal nystagmus with or without a vertical or torsional component. One patient (Pt 4) showed right and clockwise torsional (from the patient's perspective) beating nystagmus with an upbeat component that was irregularly interposed by left and counterclockwise torsional beating nystagmus with an upbeat component (aperiodic alternating nystagmus).

Direction-changing apogeotropic and geotropic nystagmus were induced during supine roll test in each patient (Pt 13 and 16, respectively). Seven patients showed ocular ipsipulsion with ipsilesional saccadic hypermetria while 11 had impaired horizontal smooth pursuit in one ($n = 4$) or both directions ($n = 7$). Accompanying neurological signs included ataxia ($n = 17$), sensory changes ($n = 11$), Horner syndrome ($n = 6$). dysphagia ($n = 5$), dysarthria ($n = 4$), and hoarseness ($n = 2$).

The head impulse VOR gains for the horizontal semicircular canals were abnormal in six patients (6/14, 43%); slightly decreased ipsilesionally ($n = 2$) or bilaterally ($n = 1$), and increased contralesionally ($n = 2$) or bilaterally ($n = 1$). Most of the patients tested exhibited ipsilesional SVV tilt (13/14, 93%).

All patients had an acute infarction involving unilateral lateral medulla in the territory of posterior inferior cerebellar artery, and five of them showed an additional infarction in the ipsilateral cerebellum (Fig. 1).

Oculographic Analyses

Slow-Phase Waveforms of Spontaneous Nystagmus

Patients showed various slow-phase waveforms of spontaneous nystagmus during visual fixation that included linear- ($n = 12$), exponentially increasing- ($n = 1$), and mixed forms ($n = 4$). The mixed-velocity waveforms mostly comprised an initial decreasing-velocity followed by a constant or increasing-velocity even in a single beat. On elimination of visual fixation, the slow phase of spontaneous nystagmus became linear in 16 patients regardless of the waveforms observed during visual fixation while the slow phase velocity remained increasing in one (Pt 17). In three patients (Pt 1, 12 and 17), the patterns of slow-phases were changed by lateral gazes in a various way (Table 2).

Table 2 Clinical and radiological characteristics of patients with spontaneous nystagmus violating Alexander's Law

Patients	Sex/age	Lesion side	Involved lesions	SN with fixation		Waveforms change during horizontal gazes	SN without fixation	
				Direction	Slow phase waveforms		Direction	Slow phase waveforms
1	F/82	L	LM, CB	R	C	D in slow-phase direction in only right eye	R=U	C
2	F/74	R	LM	L	C	(-)	L	C
3	M/72	L	LM, CB	R	mixed (D+C)	(-)	R	C
4	F/54	L	LM	R>U>CW	mixed (D+I or D+C)	(-)	R>U>CW (aPAN)	C
5	F/62	L	LM	R	mixed (D+C)	(-)	R=CW	C
6	M/31	R	LM	L>CCW>U	C	(-)	L>CCW>U	C
7	F/63	R	LM	L>CCW>U	C	(-)	L>CCW>U	C
8	F/56	L	LM	R>CW>U	mixed (D or D+I or D+C)	(-)	R>CW>U	C
9	F/87	L	LM	R>CW	C	(-)	R>CW	C
10	F/63	R	LM, CB	L>U>CCW	C	(-)	L>U>CCW	C
11	M/67	R	LM, CB	CCW>L>U	C	(-)	L>CCW>U	C
12	M/66	L	LM, CB	CW>R>U	C	mixed (D or D+C) in slow-phase direction	R>CW>U	C
13	M/61	R	LM	CCW>L>U	C	(-)	L>U>CCW	C
14	F/43	R	LM	CCW>L>U	C	(-)	L>CCW>U	C
15	M/57	R	LM	CCW>L	C	(-)	L>CCW>U	C
16	M/62	L	LM	R>CW>U	C	(-)	R=U>CW	C
17	M/52	R	LM	L=U>CCW	I	D in slow-phase direction, C in fast-phase direction	L=U>CCW	I

C constant-velocity, *CB* cerebellum, *CCW* counterclockwise, *CW* clockwise, *D* decreasing-velocity, *F* female, *I* increasing-velocity, *L* left, *LM* lateral medulla, *M* male, *aPAN* aperiodic alternating nystagmus, *R* right, *SN* spontaneous nystagmus, *U* upbeat

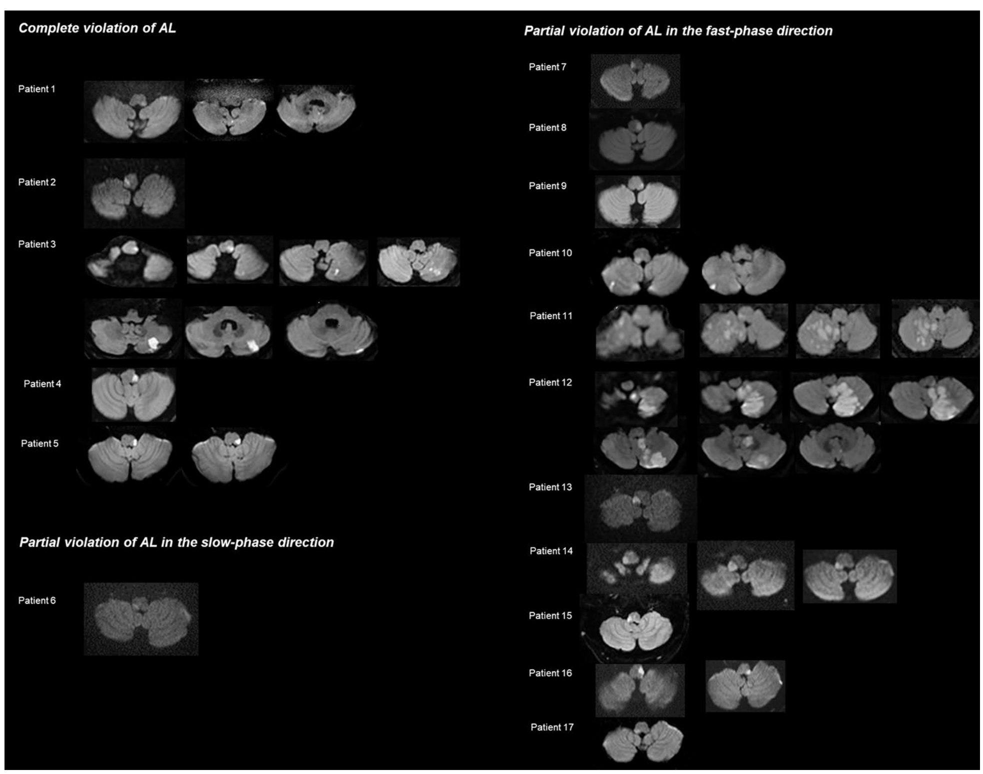

Fig. 1 Involved lesions in 17 patients with spontaneous nystagmus violating the Alexander's law. Diffusion-weighted images show the lesions mostly affecting unilateral lateral medulla. Five patients (patients 1, 3, and 10–12) have additional infarctions in the ipsilesional cerebellum

Patterns Violating the Alexander's Law

Spontaneous nystagmus violated the Alexander's law during the gazes in both directions ($n=5$, 29%) or during the gaze only in one direction ($n=12$, 71%) (Table 3). Of the five patients with a violation in both directions, one (Pt 4) showed centripetal nystagmus (beating toward the primary position during the gaze in the direction of nystagmus). All five patients showed a negative slope for gaze both in the slow- and fast-phase directions (Table 3; Fig. 2A), and the median slopes were −0.027 (IQR −0.009 ~ -0.156) and −0.103 (IQR −0.032 ~ -0.143), respectively. In these patients, spontaneous nystagmus had a linear- ($n=2$, Fig. 2A) or mixed-velocity ($n=3$) slow-phase waveforms.

Of the 12 patients with a violation of the Alexander's law only in one direction of gaze, one showed an increase of nystagmus during the gaze in the opposite direction of nystagmus (Pt 6). In this patient, the slow phase was linear, and the slope was −0.024 during the gaze in the opposite direction of spontaneous nystagmus (Table 3; Fig. 2B). In the remaining 11 patients (Pt 7–17), spontaneous nystagmus decreased when looking in the direction of spontaneous nystagmus with a median slope at -0.094 (IQR −0.033~ -0.224) (Table 3; Fig. 2C). The slow phases were mostly linear ($n=9$), but accelerating or mixed in each patient (Fig. 3).

Comparison of the Time Constant between the Controls (Patients with VN) and Patients with Spontaneous Nystagmus Violating the Alexander's Law

In the controls, the spontaneous nystagmus invariably obeyed the Alexander's law with a positive slope during the lateral gazes in both directions. The median slopes were 0.100 (IQR 0.070 ~ 0.156) during the gaze in the direction of spontaneous nystagmus and 0.099 (IQR 0.063 ~ 0.188) during the gaze in the opposite direction. The median Tc was 9.0 s (IQR 5.5 ~ 12.6 s).

In 17 patients with spontaneous nystagmus violating the Alexander's law, the median Tc was measured at 14.4 s (IQR 6.4 ~ 38.9 s) when the Alexander's law was violated and at 9.6 s (IQR 3.6 ~ 16.1 s) when the Alexander's law was obeyed. The median Tc was higher when the Alexander's was violated than that observed in the controls (14.4 s [IQR 6.4–38.9] vs. 9.0 s [IQR 5.5–12.6], $p=0.036$), whereas the Tc when the Alexander's law was obeyed in the patients did not differ from the Tc observed in the controls (9.6 s [IQR 3.6–16.1] vs. 9.0 s [IQR 5.5–12.6], $p=0.924$) (Fig. 4).

Table 3 Oculographic analysis of spontaneous nystagmus as a function of horizontal eye positions

Patient	Slow-phase direction			Primary position	Fast-phase direction		
	Velocity (mean ± SD, °/s)	Slope (°/s/°)	Tc ($\Delta E_{pos}/\Delta E_{vel}$, s)	Velocity (mean ± SD, °/s)	Velocity (mean ± SD, °/s)	Slope (°/s/°)	Tc ($\Delta E_{pos}/\Delta E_{vel}$, s)
Complete violation of AL							
1	7.24 ± 2.73	-0.012	83.3	7.00 ± 2.14	3.75 ± 1.80	-0.163	6.2
2	14.21 ± 5.14	-0.279	3.6	8.63 ± 8.21	6.58 ± 1.27	-0.103	9.8
3	2.94 ± 0.80	-0.027	37.7	2.41 ± 1.12	2.00 ± 0.82	-0.021	48.8
4	1.59 ± 0.36	-0.033	30.8	0.94 ± 0.47	-1.52 ± 0.32†	-0.123	8.1
5	1.78 ± 0.54	-0.006	166.7	1.66 ± 0.54	0.82 ± 0.37	-0.042	23.8
Partial violation of AL in the slow-phase direction							
6	4.90 ± 1.47	-0.024	42.6	4.43 ± 1.32	7.14 ± 1.76	0.136	7.4
Partial violation of AL in the fast-phase direction							
7	9.94 ± 2.06	0.085	11.8	11.63 ± 2.29	9.37 ± 2.41	-0.113	8.9
8	1.34 ± 0.23	0.027	37.7	1.87 ± 0.76	0	-0.094	10.7
9	3.20 ± 0.73	0.033	30.8	3.85 ± 0.98	2.46 ± 1.11	-0.070	14.4
10	3.29 ± 0.70	0.060	16.8	4.48 ± 2.00	0	-0.224	4.5
11	7.71 ± 3.46	0.513	2.0	17.96 ± 4.05	8.30 ± 2.44	-0.483	2.1
12	1.36 ± 0.31	0.085	11.8	3.06 ± 0.66	0	-0.153	6.5
13	-2.68 ± 1.29†	0.300	3.3	3.31 ± 1.42	2.66 ± 0.87	-0.033	30.8
14	-2.94 ± 1.15†	0.237	4.2	1.79 ± 0.72	1.15 ± 0.13	-0.032	31.3
15	-1.41 ± 0.43†	0.181	5.5	2.20 ± 0.54	1.76 ± 0.54	-0.022	45.5
16	1.39 ± 0.33	0.073	13.8	2.84 ± 0.62	1.46 ± 0.33	-0.069	14.5
17	5.19 ± 2.64	0.506	2.0	15.30 ± 2.57	9.97 ± 1.93	-0.267	3.8

† Negative value indicates that horizontal nystagmus reverses the direction during eccentric position

AL Alexander's law, *Tc* time constant, ΔE_{pos} difference of eye position between the two targets, ΔE_{vel} difference of slow-phase velocity between the two targets

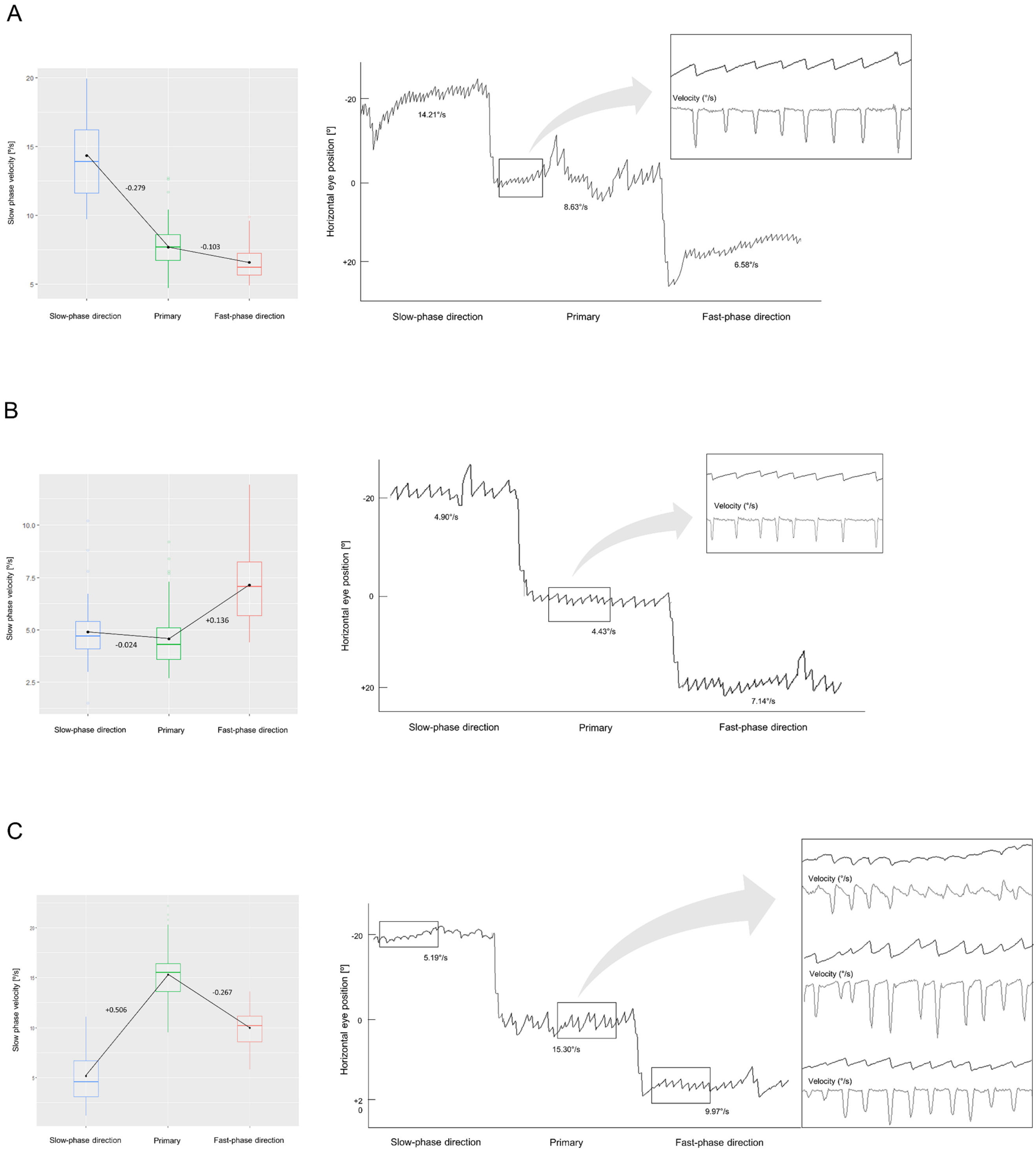

Fig. 2 The patterns violating the Alexander's law. (**A**) In patient 2, the slow-phase velocities (SPVs) of nystagmus increase when looking in the slow-phase direction while decrease when looking in the fast-phase direction. As a result, both slopes for gazes in the slow- and the fast-phase directions have negative values. The slow-phase waveforms of the nystagmus have a linear velocity during the primary and eccentric gazes (inside the rectangle). (**B**) In patient 6, the nystagmus increases during the gaze in both directions. The slow phases are linear (inside the rectangle). (**C**) In patient 17, the nystagmus decreases during the gaze in both directions, thus the slope is negative in the fast-phase direction. The slow phase waveforms are variable, showing an increasing-velocity at the primary position, and then changing into a decreasing velocity during the gaze in the slow-phase direction and a constant velocity during the gaze in the fast-phase direction (inside the rectangle)

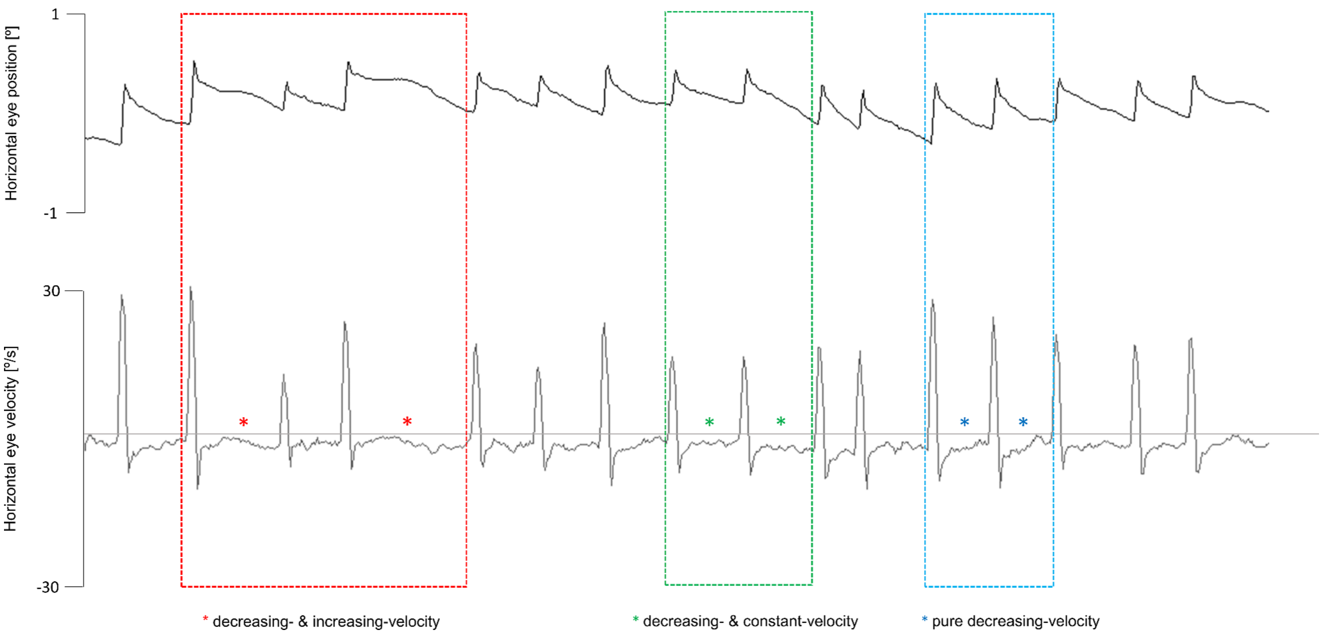

Fig. 3 The slow-phase waveform of spontaneous nystagmus with a mixed-velocity in patient 8. The slow phase starts with a rapid drift with a deceleration, which is then followed by an acceleration (red asterisks) or constant velocity (green asterisks). Some waveforms show a pure decreasing velocity (blue asterisks)

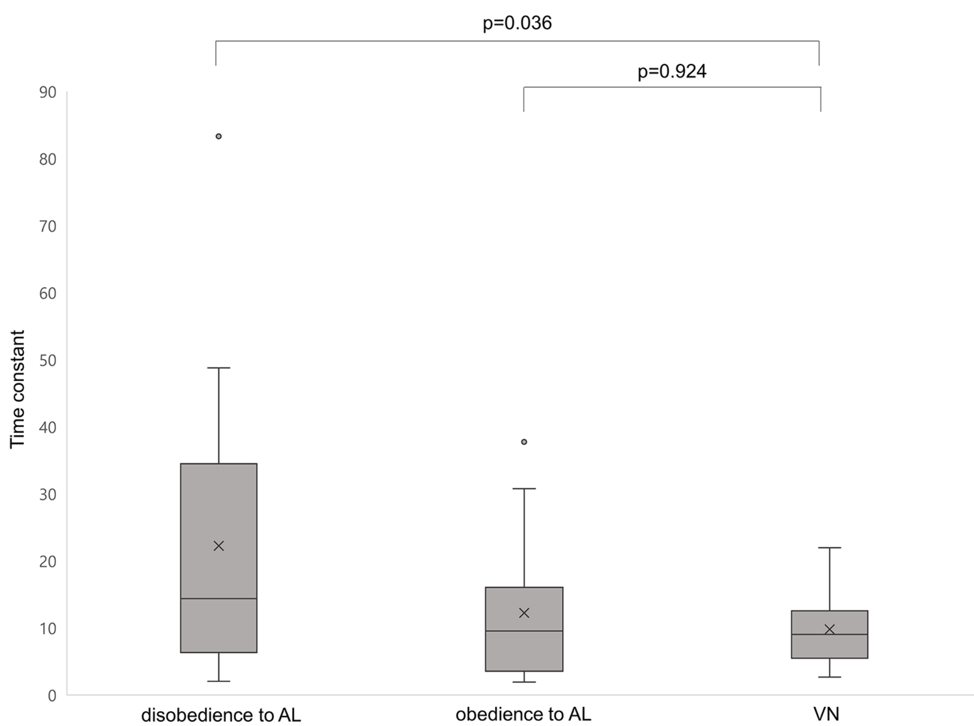

Fig. 4 Comparison of the time constants between patients with spontaneous nystagmus violating the Alexander's law and vestibular neuritis. The median Tc is higher when the Alexander's is violated than that observed in the controls (14.4 s [interquartile range, IQR 6.4–38.9] vs. 9.0 s [IQR 5.5–12.6], $p=0.036$, Mann-Whitney test), whereas the Tc when the Alexander's law is obeyed in the patients does not differ from the Tc observed in the controls (9.6 s [IQR 3.6–16.1] vs. 9.0 s [IQR 5.5–12.6], $p=0.924$). AL = Alexander's law, VN = vestibular neuritis

Modeling

Neural Integration Model

To test the study hypothesis that an unstable neural integrator can generate nystagmus violating Alexander's law, we primarily utilized the gaze-holding neural integrator model [13], incorporating lesion-induced changes. Based on the neural activities being proportional to eye position, the burst-tonic neurons within the vestibular nucleus and nucleus prepositus hypoglossi are believed to be the primary neural integrator, converting eye velocity into position information. However, this process is not perfect and is known as 'leakiness,' which might offer advantages in

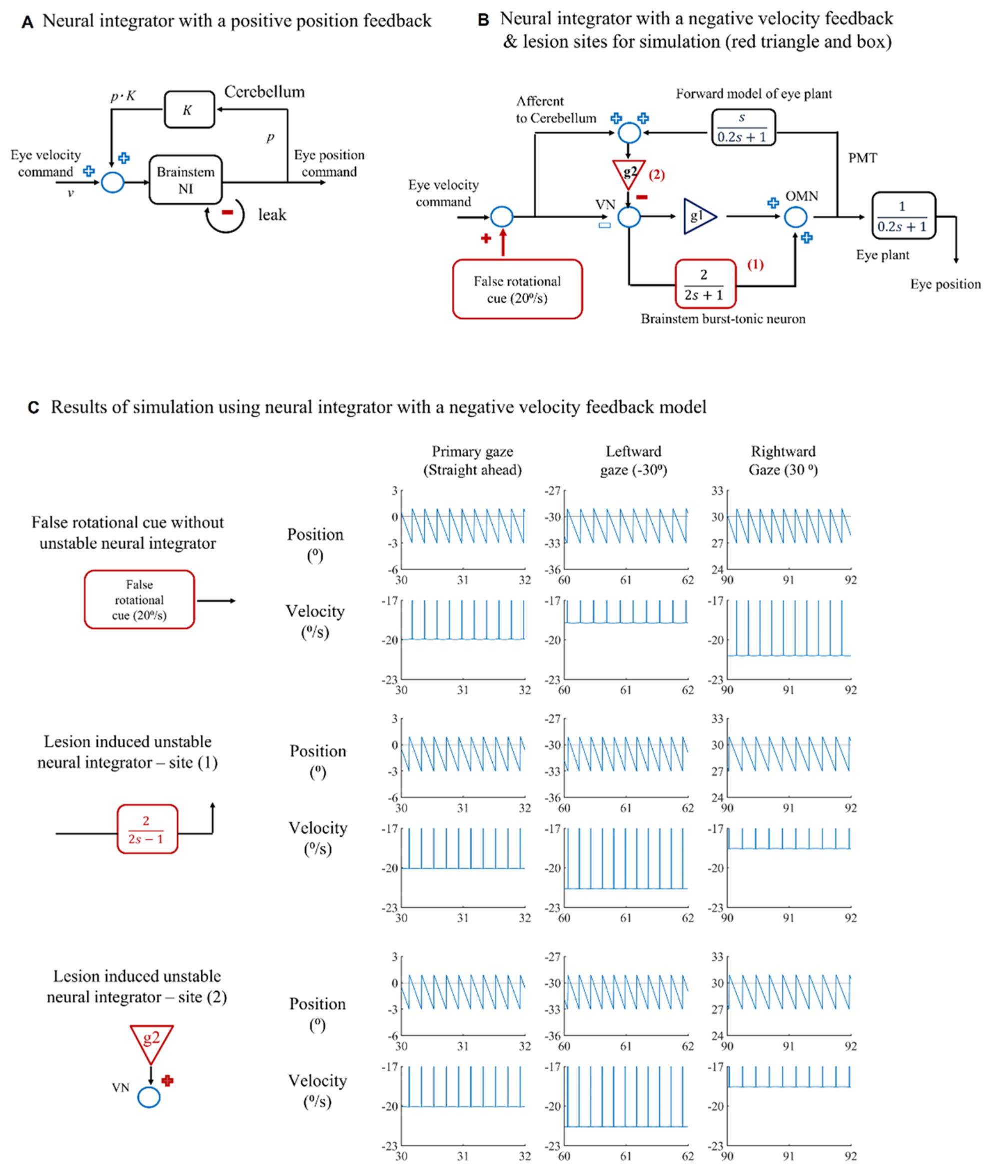

Fig. 5 Schematics of neural integrator models and the result of simulations. (**A**) In the positive feedback model, the cerebellum compensates for the leakiness of the neural integrator with position information. (**B**) The model of negative velocity feedback. The vestibular signals primarily reach the vestibular nucleus (VN). These signals are then sent to the ocular motor nucleus (OMN) via both the direct and the neural integrator pathways, which in turn send signals to the eye plant. The cerebellum receives predicted eye velocity signals using a forward model of the eye plant, as well as desired eye velocity information through primary vestibular afferents. The difference between these signals is sent to the negative feedback pathway. (**C**) For the simulation, three changes were made. One is the introduction of a false rotational cue (20 degrees/s clockwise in the yaw axis), simulating lesion-induced vestibular bias. This results in spontaneous nystagmus directed rightward. With normal integrator function, the gaze direction during the quick phase of nystagmus increases eye velocity, while gaze in the opposite direction decreases it. However, lesions at sites 1 and 2 alter the function of the neural integrator, making it unstable. These changes generate spontaneous nystagmus that does not comply with Alexander's law during lateral gaze movements

counteracting the accumulation of biological noise. This leakiness was herein modeled as a low-pass filter with a time constant of 2 s [13]. The flocculus and paraflocculus then play an important role in compensating for the leakiness of the brainstem neural integrator. They can be modeled either through positive feedback or negative feedback. In the positive feedback model, the eye position signals generated in the brainstem neural integrator are fed back to the neural integrator itself via the pathway through the cerebellum (Fig. 5A) [2, 14]. Meanwhile, the negative feedback model provides feedback to the vestibular nucleus and uses velocity information based on the difference between predicted and estimated eye velocities (Fig. 5B) [13]. The details of each model have been introduced elsewhere [13, 14], but herein we adopted the negative feedback model as it is considered more physiological and helps stabilize the response of the system.

Lesion-Induced Changes

Because the lesions observed for the violation of Alexander's law were commonly located in the lateral medulla, these lesions could directly affect the brainstem neural integrator itself or the neural synapse between Purkinje cells and the brainstem vestibular nucleus. Therefore, we set the lesion-induced change such that a hyperexcitable brainstem neural integrator abnormally accumulates neural signals, replacing leakiness, or the cerebellar input exerts a positive, instead of a negative, effect on the vestibular nucleus (site-1 in Fig. 5B and C). The basis of these changes is as follows: For the direct neural integrator lesions, we drew inspiration from the Hodgkin and Huxley cell model [15], which describes that neurons transform dendritic inputs into action potentials using voltage-gated Na^+ and K^+ channels and leak, thereby approximating neuronal behavior as a leaky integrator acting as a low-pass filter. The inherent leak in neurons mainly relies on non-voltage-gated ion channels, which allow passive ion flow driven by electrochemical gradients, thereby stabilizing the cell at its resting membrane potential. Consequently, lesion-induced changes in membrane integrity or ion channel function can disrupt the leaky component of the cell membrane, leading to instability in neural firing patterns in response to synaptic inputs. Mathematically, this shift can be represented by modifying the transfer function of brainstem neural integrator from $\frac{2}{2s+1}$ to $\frac{2}{2s-1}$ (site-1 in Fig. 5B and C). For cerebellar positive feedback, it has been suggested that GABA, the inhibitory neurotransmitter utilized by Purkinje cells, can paradoxically exert an excitatory effect (site-2 in Fig. 5B and C). For instance, after neuronal injury, dysfunction of cation-chloride cotransporters can elevate intracellular Cl^- levels [16]. Under such conditions, GABA receptor activation paradoxically causes Cl^- to flow out of the cell, leading to depolarization instead of hyperpolarization [16] Additionally, Purkinje neurons, which normally suppress the activity of target neurons through asynchronous firing, have been shown to paradoxically facilitate the activity of target neurons when firing synchronously [17]. Lastly, we implemented false constant rotational cues to induce lesion-induced spontaneous nystagmus.

Simulation

The results show that with normal integrator functions, the false rotational cue generates nystagmus following Alexander's law (first row in Fig. 5C). The first lesion, which changes the brainstem neural integrator, and the second lesion, which causes the Purkinje synapse to exert excitatory input, both lead to nystagmus that violates Alexander's law (second and third rows in Fig. 5C). This successfully visualizes our hypothesis.

Discussion

This study demonstrates that violation of the Alexander's law is an occasional finding in patients with Wallenberg syndrome and an unstable neural integrator would be an underlying mechanism of the phenomenon. We provide evidence to support this hypothesis using a gaze-holding neural integrator model that incorporates lesion-induced changes.

An explanation for Alexander's law is adaptive changes in the velocity-to-position neural integrator to reduce spontaneous vestibular nystagmus during the gaze in the slow-phase direction to bring about improvement of vision [2–5, 18]. The neural integrator is responsible for holding the eyes in eccentric position by mathematically integrating the eye velocity commands [2, 19, 20]. Impaired gaze holding and resultant gaze-evoked nystagmus (GEN) may be observed in lesions involving the nucleus prepositus hypoglossi (NPH) and medial vestibular nucleus (MVN) that play a crucial role in horizontal neural integration [21–24]. If the neural integrator becomes leaky in patients with direction-fixed vestibular nystagmus, the drift velocity of nystagmus increases during the gaze in the fast-phase direction and decreases during gaze in the slow-phase direction, conforming to the Alexander's law.

When the neural integrator becomes unstable, the eyes drift away from the central position, causing centripetal nystagmus [2, 25–29]. Theoretically, when an unstable neural integrator is combined with direction-fixed vestibular nystagmus, the drift velocity may increase during the gaze in the slow-phase direction and decrease during the gaze in the fast-phase direction, thereby violating the Alexander's law. In support of this hypothesis, our patients with Wallenberg syndrome and violation of the Alexander's law showed that the Tc of nystagmus is increased only during the gaze where the Alexander's law was disobeyed. In addition, some patients exhibited the nystagmus with an accelerating (increasing velocity) slow phase or centripetal nystagmus. Our patients with the violation of Alexander's law had invariable lesions in the lateral medulla. These lesions could directly affect the brainstem neural integrator itself or the neural synapse between Purkinje cells and the brainstem vestibular nucleus [30, 31]. The neural integration for horizontal eye movements depends on a distributed network of neurons lying in the brainstem and cerebellum [2, 21–24]. The flocculus and paraflocculus enhance the performance of an inherently leaky neural integrator in the brainstem either through positive or negative feedback [13, 14, 32, 33]. Our gaze-holding neural integrator model, incorporating lesion-induced changes showed that when the neural integrator becomes unstable, nystagmus violates Alexander's law. Under normal integrator function, the false rotational cue generates nystagmus following Alexander's law. In contrast, the lesion-induced change such that a hyperexcitable brainstem neural integrator abnormally accumulates neural signals, replacing leakiness, or that the cerebellar input exerts a positive, instead of a negative, effect on the vestibular nucleus, produces nystagmus that violates Alexander's law. Increased intracellular Ca2+concentration by lesion-induced changes in neuronal membrane integrity or ion channel function may cause neuronal hyperexcitability for the direct neural integrator lesions [15]. For the cerebellar positive feedback, it has been suggested that GABA, the inhibitory neurotransmitter utilized by Purkinje cells, may paradoxically exert an excitatory effect after neuronal injury by causing intracellular Cl^- to flow out of the cell [16]. Alternatively, the lesion may facilitate the activity of target neurons in the cerebellar nuclei by synchronous firing of Purkinje cells, leading to the unstable neural integrator [17]. Consistent with our results, previous experimental studies also showed that injection of either bicuculline or muscimol into the MVN caused instability of gaze holding, in which the eye drifts away from the central position with increasing-velocity waveforms, implying an unstable neural integrator [34, 35]. These effects may be related to inactivation either of neurons within NPH-MVN or the cerebellar projections to them that control the fidelity of neural integration.

Alternatively, the caudal cell groups of the paramedian tract (PMT) may be another neural substrate for nystagmus violating Alexander's law in the medulla. The medullary PMT cells are also involved in the gaze holding by providing a motor-feedback signal for eye position and velocity to the flocculus [2]. Thus, the PMT lesion may cause an unstable neural integrator for gaze and resultant offset in the null position of the integrator, which leads to violate Alexander's law. Previous reports have also implicated dysfunction of PMT cells to cause inverse eye position dependency of upbeat or downbeat nystagmus in medullary lesions [36, 37]. In our patients, however, the lesions mostly did not invade the paramedian region of the medulla where the PMT cells are located, suggesting that a dysfunction of PMT cells is unlikely to be a cause of unstable horizontal neural integrator.

Patients with a lesion restricted to the vestibular nuclei may show diverse signs of both peripheral vestibulopathy (spontaneous nystagmus, caloric paresis, and positive head impulse tests) and central vestibular dysfunction (GEN) [31, 38–40]. Our study disclosed that lesions in the area of the vestibular nuclei can lead to diverse patterns of spontaneous nystagmus and its modulation by visual fixation and gaze. These findings may be explained by significant divergence of excitatory projections from the peripheral vestibular structures and brainstem neural integrators, along with inhibitory inputs from the cerebellum [2, 13, 30, 40].

Some patients exhibited a mixed pattern of slow-phase velocity of spontaneous nystagmus with a beat-to-beat and intrabeat variability. Even in a single beat, the nystagmus consisted of an initial decelerating and following linear or accelerating slow phases. Similar waveforms have been reported in vertical nystagmus observed in lesions involving the PMT [41] or in patients with ankylosing spondylitis [42]. According to a mathematical model, these unusual waveforms may occur due to the pulse-step mismatch creating the initially decreasing-velocity waveforms due to leaky neural integrators, and subsequent disruption of the cerebellar feedback for gaze holding through PMT, resulting in unstable neural integrators and nystagmus with an increasing-velocity waveform [41, 42]. GEN and superimposed pendular nystagmus observed in patients with multiple sclerosis also imply that the neural integrator can be simultaneously leaky and unstable [43].

Our patients more frequently showed a violation of the Alexander's law only in one direction. This indicates that dysfunction of the neural integration depends on gaze direction. Thus, the neural integrators may become unstable in one direction while they become leaky in other directions of gaze [44–46]. Indeed, asymmetrical GEN has been described in patients with unilateral lesions of the NPH or MVN [31, 38–40, 47]. In unilateral lesion of the NPH, the neural integration is more severely impaired with a decrease in the Tc of postsaccadic drift after ipsilesional eccentric gaze [47].

Conclusion

Spontaneous nystagmus in patients with Wallenberg's syndrome can violate Alexander's law, which permit a differentiation from acute peripheral vestibular syndrome. The violation of the Alexander's law may be attributed to unstable neural integrator and can be simulated by lesions affecting the brainstem neural integrator itself or the neural synapse between Purkinje cells and the brainstem vestibular nucleus.

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Author Contributions J-HC and E-HO analyzed and interpreted the data and wrote the manuscript. H-SK, J-YP, S-ML, S-YC, H-JK, J-YC, J-SK and JOM analyzed and interpreted the data. K-DC designed and conceptualized the study, interpreted the data, and revised the manuscript. All authors reviewed and approved the final version of the manuscript. K-DC is responsible for the overall content as guarantor, and obtained funding.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Ethical Approval All the experiments performed in this study followed the tenets of the Declaration of Helsinki. Informed consents were obtained from the participants (including technicians in supplementary video) after the nature and possible consequences of this study had been explained to them. This study was approved by the Institutional Review Board of Pusan National University (2308-031-130) and Pusan National University Yangsan Hospitals (05-2023-180).

Competing Interests The authors declare no competing interests.

References

1. Alexander G. Die Ohrenkrankheiten im Kindesalter. In: Handbuch der Kinderheilkunde, edited by Schlossmann A. Leipzig, Germany: Vogel, 1912, pp. 84–96.
2. Leigh RJ, Zee DS. The neurology of Eye Movements. 5th ed. New York: Oxford University Press; 2015.
3. Hegemann S, Straumann D, Bockisch C. Alexander's law in patients with acute vestibular tone asymmetry–evidence for multiple horizontal neural integrators. J Assoc Res Otolaryngol. 2007;8:551–61.
4. Robinson DA, Zee DS, Hain TC, Holmes A, Rosenberg LF. Alexander's law: its behavior and origin in the human vestibulo-ocular reflex. Ann Neurol. 1984;16:714–22.
5. Khojasteh E, Bockisch CJ, Straumann D, Hegemann SC. A mechanism for eye position effects on spontaneous nystagmus. Annu Int Conf IEEE Eng Med Biol Soc. 2012;2012:3572–5.
6. Doslak MJ, Dell'Osso LF, Daroff RB. A model of Alexander's law of vestibular nystagmus. Biol Cybern. 1979;34:181–6.
7. Bockisch CJ, Khojasteh E, Straumann D, Hegemann SC. Development of eye position dependency of slow phase velocity during caloric stimulation. PLoS ONE. 2012;7:e51409.
8. Shin BS, Oh SY, Kim JS, Lee H, Kim EJ, Hwang SB. Upbeat nystagmus changes to downbeat nystagmus with upward gaze in a patient with Wernicke's encephalopathy. J Neurol Sci. 2010;298:145–7.
9. Newman-Toker DE, Kerber KA, Hsieh YH, Pula JH, Omron R, Saber Tehrani AS, et al. HINTS outperforms ABCD2 to screen for stroke in acute continuous vertigo and dizziness. Acad Emerg Med. 2013;20:986–96.
10. Mantokoudis G, Korda A, Zee DS, Zamaro E, Sauter TC, Wagner F, et al. Bruns' nystagmus revisited: a sign of stroke in patients with the acute vestibular syndrome. Eur J Neurol. 2021;28:2971–9.
11. Lädrach C, Zee DS, Wyss T, Wimmer W, Korda A, Salmina C, et al. Alexander's Law during High-Speed, Yaw-Axis Rotation: adaptation or saturation? Front Neurol. 2020;11:604502.
12. Choi JH, Oh EH, Choi SY, Kim HJ, Lee SK, Choi JY, et al. Vestibular impairments in episodic ataxia type 2. J Neurol. 2022;269:2687–95.
13. Glasauer S. Cerebellar contribution to saccades and gaze holding: a modeling approach. Ann N Y Acad Sci. 2003;1004:206–19.
14. Zee DS, Leigh RJ, Mathieu-Millaire F. Cerebellar control of ocular gaze stability. Ann Neurol. 1980;7:37–40.
15. HODGKIN AL, HUXLEY AF. A quantitative description of membrane current and its application to conduction and excitation in nerve. J Physiol. 1952;117:500–44.
16. Kaila K, Price TJ, Payne JA, Puskarjov M, Voipio J. Cation-chloride cotransporters in neuronal development, plasticity and disease. Nat Rev Neurosci. 2014;15:637–54.
17. Person AL, Raman IM. Purkinje neuron synchrony elicits time-locked spiking in the cerebellar nuclei. Nature. 2011;481:502–5.
18. Bockisch CJ, Hegemann S. Alexander's law and the oculomotor neural integrator: three-dimensional eye velocity in patients with an acute vestibular asymmetry. J Neurophysiol. 2008;100:3105–16.
19. Godaux E, Cheron G. The hypothesis of the uniqueness of the oculomotor neural integrator: direct experimental evidence in the cat. J Physiol. 1996;492:517–27.
20. Arnold DB, Robinson DA. The oculomotor integrator: testing of a neural network model. Exp Brain Res. 1997;113:57–74.
21. Dale A, Cullen KE. The nucleus prepositus predominantly outputs eye movement-related information during passive and active self-motion. J Neurophysiol. 2013;109:1900–11.
22. McConville K, Tomlinson RD, King WM, Paige G, Na EQ. Eye position signals in the vestibular nuclei: consequences for models of integrator function. J Vestib Res. 1994;4:391–400.
23. McFarland JL, Fuchs AF. Discharge patterns in nucleus prepositus hypoglossi and adjacent medial vestibular nucleus during horizontal eye movement in behaving macaques. J Neurophysiol. 1992;68:319–32.
24. Sylvestre PA, Choi JT, Cullen KE. Discharge dynamics of oculomotor neural integrator neurons during conjugate and disjunctive saccades and fixation. J Neurophysiol. 2003;90:739–54.
25. Leech J, Gresty M, Hess K, Rudge P. Gaze failure, drifting eye movements, and centripetal nystagmus in cerebellar disease. Br J Ophthalmol. 1977;61:774–881.
26. Barton JJ, Sharpe JA. Oscillopsia and horizontal nystagmus with accelerating slow phases following lumbar puncture in the Arnold-Chiari malformation. Ann Neurol. 1993;33:418–21.
27. Leigh RJ, Thurston SE, Tomsak RL, Grossman GE, Lanska DJ. Effect of monocular visual loss upon stability of gaze. Invest Ophthalmol Vis Sci. 1989;30:288–92.
28. Leigh RJ, Zee DS. Eye movements of the blind. Invest Ophthalmol Vis Sci. 1980;19:328–31.
29. Schneider RM, Thurtell MJ, Eisele S, Lincoff N, Bala E, Leigh RJ. Neurological basis for eye movements of the blind. PLoS ONE. 2013;8:e56556.
30. Choi KD, Oh SY, Park SH, Kim HJ, Koo JW, Kim JS. Head-shaking nystagmus in lateral medullary infarction: patterns and possible mechanisms. Neurology. 2007;68:1337–44.
31. Lee SU, Park SH, Park JJ, Kim HJ, Han MK, Bae HJ, et al. Dorsal medullary infarction: distinct syndrome of isolated Central Vestibulopathy. Stroke. 2015;46:3081–7.
32. Kamath RY, Keller EL. A neurological integrator for the oculomotor control system. Math Biosci. 1976;30:341–52.
33. Beh SC, Frohman TC, Frohman EM. Cerebellar control of Eye Movements. J Neuroophthalmol. 2017;37:87–98.
34. Straube A, Kurzan R, Büttner U. Differential effects of bicuculline and muscimol microinjections into the vestibular nuclei on simian eye movements. Exp Brain Res. 1991;86:347–58.
35. Arnold DB, Robinson DA, Leigh RJ. Nystagmus induced by pharmacological inactivation of the brainstem ocular motor integrator in monkey. Vis Res. 1999;39:4286–95.
36. Helmchen C, Glasauer S, Sprenger A. Inverse eye position dependency of downbeat nystagmus in midline medullary lesion. J Neurol. 2013;260:2908–10.
37. Hyun JW, Choi SY, Huh YE, Kwon OK, Kim JS. Upbeat nystagmus due to a giant vertebral artery aneurysm. Neurol Sci. 2013;34:805–7.
38. Kim HJ, Lee SH, Park JH, Choi JY, Kim JS. Isolated vestibular nuclear infarction: report of two cases and review of the literature. J Neurol. 2014;261:121–9.
39. Ataç C, Kısabay A, Çetin Akkoç C, Saruhan G, Çelebisoy N. Vestibular nuclear infarction: Case series and review of the literature. J Stroke Cerebrovasc Dis. 2020;29:104937.
40. Kim HS, Choi JH, Oh EH, Choi SY, Choi KD. Dorsolateral pontine lesions produce distinct ocular motor abnormalities with anatomical correlations. Eur J Neurol. 2025;32(1):e70010.
41. Tian J, Otero-Millan J, Zee DS, Kheradmand A. Upbeat nystagmus with an unusual velocity-decreasing and increasing waveform: a sign of Gaze-Holding Dysfunction in the Paramedian tracts in the Medulla? Cerebellum. 2023;22:148–54.

42. Abel LA, Traccis S, Dell'Osso LF, Ansevin CF. Variable waveforms in downbeat nystagmus imply short-term gain changes. Ann Neurol. 1983;13:616–20.
43. Gupta P, Shaikh AG. Leaky and unstable neural integrator can coexist-Paradox observed in multiple sclerosis. J Neuroophthalmol. 2020;40:226–33.
44. Cannon SC, Robinson DA, Shamma S. A proposed neural network for the integrator of the oculomotor system. Biol Cybern. 1983;49:127–36.
45. Crawford JD, Vilis T. Modularity and parallel processing in the oculomotor integrator. Exp Brain Res. 1993;96:443–56.
46. Aksay E, Olasagasti I, Mensh BD, Baker R, Goldman MS, Tank DW. Functional dissection of circuitry in a neural integrator. Nat Neurosci. 2007;10:494–504.
47. Kim SH, Zee DS, du Lac S, Kim HJ, Kim JS. Nucleus prepositus hypoglossi lesions produce a unique ocular motor syndrome. Neurology. 2016;87:2026–33.